

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Operating Systems

COURSE CODE: CSA-0716

COURSE OUTCOME COVERED

CO3: Evaluate memory management mechanisms such as linking, paging, and segmentation to determine their effectiveness for different system requirements

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks: 50

Experiment Questions:

- 1.Design and implement a program to simulate the translation of a logical address into a physical address using base and limit registers. The program should accept logical address as input, check for validity, and compute the corresponding physical address. Display appropriate messages for valid and invalid addresses.
- 2.Write a program to demonstrate the paging memory management scheme. The program should divide memory into fixed-size pages and frames, accept page number and offset as input, and compute the corresponding physical address using a page table.
- 3.Develop a program to simulate a page table. The program should allow the user to input page table entries and logical addresses, then map them to physical addresses. Clearly show how page numbers are translated into frame numbers.
- 4.Implement a program to simulate the working of a Translation Lookaside Buffer (TLB). The program should demonstrate how TLB improves memory access time by calculating the effective memory access time with and without TLB.
- 5.Write a program to illustrate memory protection using base and limit registers. The program should verify whether a given logical address lies within the valid range and prevent unauthorized access.

ANSWERS

EXPERIMENT-1: Logical Address to Physical Address using Base and Limit Registers

Aim:

To simulate logical to physical address translation using **Base** and **Limit** registers.

Algorithm:

1. Read Base Register value.
2. Read Limit Register value.
3. Read Logical Address.
4. If Logical Address < Limit
 - o Physical Address = Base + Logical Address.
5. Otherwise display Invalid Address.

Sample Calculation:

Given:

- Base Register = **1000**
- Limit Register = **500**
- Logical Address = **250**

Since

$250 < 500$ (Valid)

Physical Address

$= 1000 + 250$

= 1250

Sample Output:

Base Register = 1000

Limit Register = 500

Logical Address = 250

Physical Address = 1250

Invalid Case:

Logical Address = 600

$600 > 500$

Output:

Invalid Logical Address

Memory Protection Fault

Result:

Logical address is successfully translated into physical address using Base and Limit registers.

EXPERIMENT-2: Paging Memory Management Scheme

Aim:

To implement paging and convert logical address into physical address.

Algorithm:

1. Read Page Size.
2. Read Page Table entries.
3. Read Page Number and Offset.
4. Find corresponding Frame Number.
5. Physical Address

= Frame Number $\times$ Page Size + Offset.

Sample Calculation:

Page Size = **1024 bytes**

Page Table

Page	Frame
0	5
1	8
2	2
3	6

Logical Address

Page Number = **2**

Offset = **500**

Frame Number = **2**

Physical Address

= $(2 \times 1024) + 500$

= $2048 + 500$

= **2548**

Sample Output:

Page Number = 2

Offset = 500

Frame Number = 2

Physical Address = 2548

Result:

Logical address is translated correctly using paging memory management.

EXPERIMENT-3: Page Table Simulation**Aim:**

To simulate page table mapping between pages and frames.

Algorithm:

1. Read number of pages.
2. Read Frame Number for every page.
3. Read Logical Address.
4. Extract Page Number and Offset.
5. Obtain Frame Number.
6. Calculate Physical Address.

Sample Calculation:

Page Size = **1024**

Page Table

Page	Frame
0	4
1	7
2	1
3	5

Logical Address = **2600**

Page Number

$2600 \div 1024$

= **2**

Offset

$2600 \% 1024$

= **552**

Frame Number

= **1**

Physical Address

$= (1 \times 1024) + 552$

= **1576**

Sample Output:

Logical Address = 2600

Page Number = 2

Offset = 552

Frame Number = 1

Physical Address = 1576

Result

The page table successfully maps logical addresses into physical addresses.

EXPERIMENT-4: Translation Lookaside Buffer (TLB)

Aim:

To simulate TLB and calculate Effective Memory Access Time (EMAT).

Algorithm:

1. Read TLB Hit Ratio.
2. Read TLB Access Time.
3. Read Memory Access Time.
4. Calculate

EMAT

$= h(TLB + M) + (1-h)(TLB + 2M)$

where

h = Hit Ratio.

Sample Calculation:

Given

- TLB Access Time = **10 ns**
- Memory Access Time = **100 ns**

- Hit Ratio = **80% = 0.8**
- Without TLB
- $= 2 \times 100$
- **= 200 ns**
- With TLB
- $= 0.8(10 + 100) + 0.2(10 + 200)$
- $= 0.8 \times 110 + 0.2 \times 210$
- $= 88 + 42$
- **= 130 ns**

Sample Output:

TLB Hit Ratio = 80%

Memory Access Time without TLB = 200 ns

Effective Memory Access Time with TLB = 130 ns

Result

TLB reduces memory access time from **200 ns** to **130 ns**, improving system performance.

EXPERIMENT-5: Memory Protection using Base and Limit Registers

Aim:

To verify memory protection using Base and Limit registers.

Algorithm:

1. Read Base Register.
2. Read Limit Register.
3. Read Logical Address.
4. If Logical Address < Limit
 - Allow access.
5. Else
 - Deny access.

Sample Calculation”:

Base = **5000**

Limit = **1000**

Logical Address = **700**

Since

$700 < 1000$

Physical Address

= $5000 + 700$

= **5700**

Access Granted.

Invalid Case:

Logical Address = **1200**

$1200 > 1000$

Access Denied.

Sample Output:

Base Register = 5000

Limit Register = 1000

Logical Address = 700

Physical Address = 5700

Access Granted

Invalid Output:

Logical Address = 1200

Access Denied

Memory Protection Violation

Result

The program correctly protects memory by allowing only valid logical addresses and blocking unauthorized access.

RUBRICS FOR CALCULATION:

Criteria	Weightage (Marks)	Level 4 – Exemplary (Full Marks)	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	35 Marks	Demonstrates clear, complete, and in-depth understanding of OS concepts (types, structure, system calls)	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answers	25 Marks	All answers are correct, precise, and relevant	Mostly correct with minor errors	Partially correct with noticeable errors	Mostly incorrect or irrelevant
Application of Knowledge	15 Marks	Effectively applies concepts to scenarios/questions	Applies with minor mistakes	Limited or partial application	No meaningful application
Completeness of Response	15 Marks	All parts of each question are fully answered	Most parts covered with small omissions	Some parts missing	Incomplete answers
Clarity & Presentation	10 Marks	Clear, well-structured, and neatly presented answers	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand